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Some Paleomagnetic Constraints on the Tectonic Evolution of the Coastal Cordillera of Central Chile

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Paleomagnetic samples collected from 19 sites in radioisotopically dated Jurassic mafic dykes and plutons in the coastal cordillera of Chile between latitude 32°15' and 33°30' S yield paleomagnetic poles (calculated by averaging the virtual geomagnetic poles from approximately coeval sites) at latitude 85.9°S, longitude 36.7°E (N = 8, $a95$ = 11.9°) for sites dated at close to 170 Ma and at latitude 80.3°S, longitude 207.8°E (N = 10, $a95$ = 9.5°) for sites dated at close to 145 Ma. These paleomagnetic poles from the coast of Chile coincide with poles from coeval sequences within the South American craton. Some of the igneous rocks that constitute these sampling sites crosscut a high-grade metamorphic complex that is distinct from other segments of the coastal cordillera in age, structural style, and metamorphic grade. This complex cannot be correlated with metamorphic rocks anywhere else in central western South America and could be considered a potentially "exotic" allochthonous terrane. If this terrane has been added to the western South American margin, then accretion occurred prior to 170 Ma. Paleomagnetic results presented here indicate that no major rotation or large-scale latitudinal translation of crustal blocks has taken place in this part of central Chile since Middle Jurassic time.

INTRODUCTION

During the last 15 years it has become apparent that the western margin of North America has grown and that the process of growth was related to the subduction of oceanic lithosphere [e.g., *Monger et al.*, 1972; *Coney et al.*, 1980]. Paleomagnetic studies have proven to be a powerful tool in understanding the history of continental growth and evolution, particularly in areas where plate convergence and/or transform faulting have caused accretion, large-scale latitudinal transport, and rotation of geologic terranes [cf. *Alvarez et al.*, 1980; *Beck*, 1976, 1980; *Irving et al.*, 1980; *McCabe*, 1984]. East-directed subduction has been taking place along the central western South American margin since at least the Triassic [*James*, 1971], and hence by analogy with western North America and southeast Asia, it is possible that accretion and transport of allochthonous terranes may have occurred here [*Nur and Ben-Avraham*, 1981]. It has also been suggested that a relation exists between the relative motion of the interacting plates in a subduction system and the tectonics of the upper plate [*Beck*, 1983], but this theory remains largely untested. The western margin of South America has a long history of dynamic plate interaction: study of the crust of coastal Chile offers great promise of improving our understanding of the relationship between plate motions and continental tectonics [e.g., *Dalziel and Forsythe*, 1985]. The high-grade metamorphic complex described below is distinct from other segments of the coastal cordillera of Chile in its age, structural style, and metamorphic grade and cannot be correlated with metamorphic rocks anywhere else in central western South America (F. Hervé, personal communication 1983): it could therefore be considered a potentially "exotic" accreted terrane [*Nur and Ben-Avraham*, 1978]. Paleomagnetic and radioisotopic data for mafic dykes and plutons that intrude this metamorphic terrane are reported here, however, and indicate that no significant rotation or translation of this part of the Chilean coastal cordillera, relative to the South American craton, has taken place since Middle Jurassic time.

GEOLOGIC BACKGROUND

Between latitude 33°45'S and 32°10'S, the coastal cordillera of Chile comprises a variety of rock types that for the most part are little studied and sparsely dated. This part of coastal Chile is included within a late Paleozoic forearc province by *Hervé et al.* [1981] and *Forsythe* [1983]. This region can be divided into three major geologic units (Figure 1).

1. Between latitude 33°42' and 32°58'S occurs a high-

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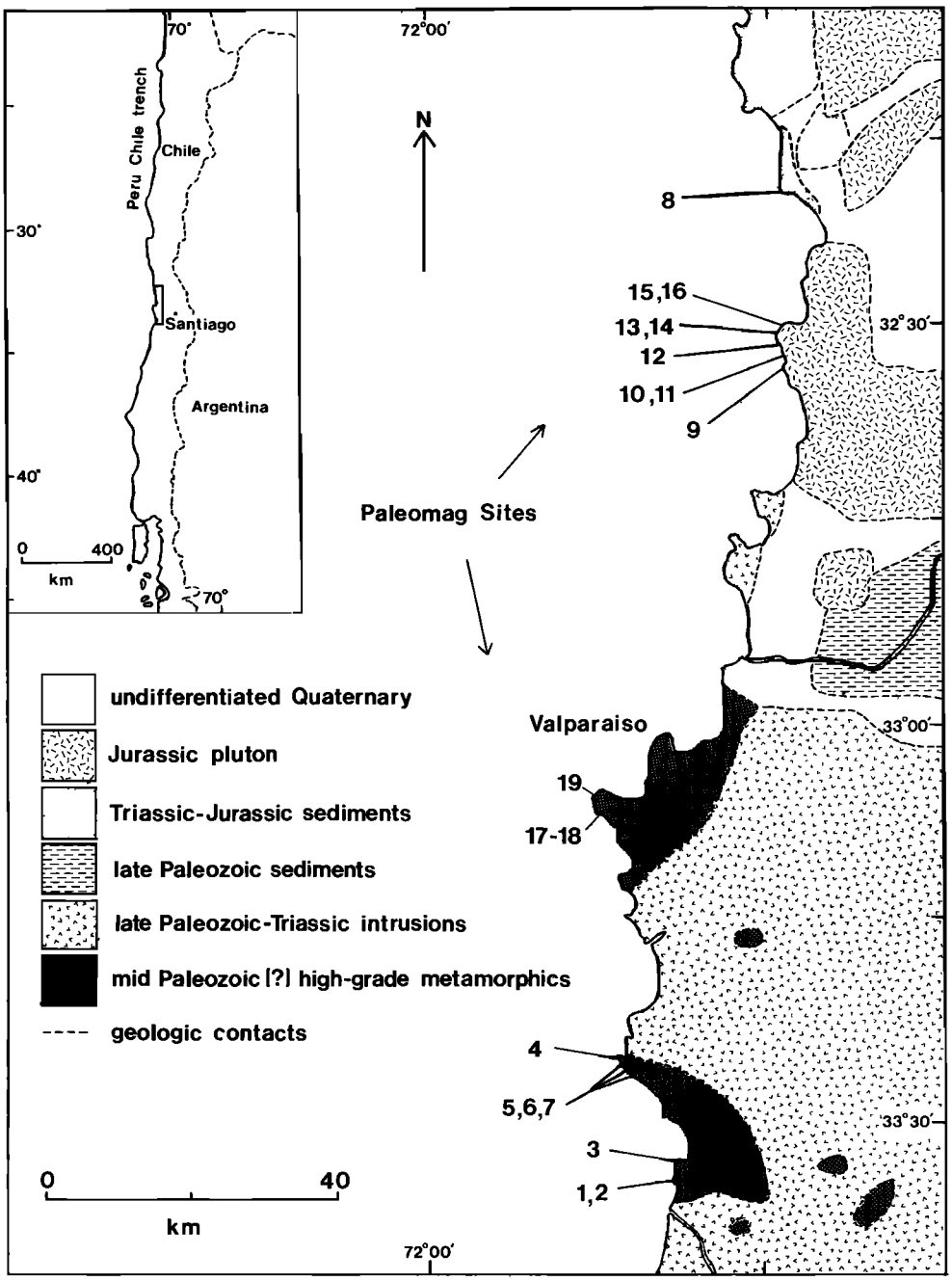


Fig. 1. Location map and geologic map showing paleomagnetic sampling site locations.

grade metamorphic and plutonic complex that is composed of at least five distinct rock types. (1) Gneiss, for the most part a high-grade metasedimentary sequence, of probable mid-Paleozoic age on the basis of U-Pb zircon data [*Corvalan and Munizaga*, 1972]. These high-grade metamorphic rocks are characterized by tight to isoclinal, upright folds in compositional layering [*Hervé*, 1976]. This structural style and grade of metamorphism is not observed elsewhere in the coastal cordillera of central Chile, which consists for the most part of low-grade metamorphic rocks (summarized by *Aguirre et al.* [1972] and *Hervé et al.* [1981]). (2) Medium-grained, equigranular granite of late Paleozoic age [*Caminos et al.*, 1982; *Drake et al.*, 1982]. (3) Medium-grained, equigranular quartz diorite, dated at between late Paleozoic and lower Triassic. (4) Medium-grained amphibolites. (5) Mafic dykes and a fine-grained intrusion that crosscut the above-mentioned units and which are part of this study (sites 1—7, 17, 18, 19).

2. Between latitude 32°25'and 32°45'S occurs a quartz diorite to gabbroic pluton (sites 10, 11, 12, 16) that crosscuts unit 1 above and is in turn crosscut by mafic dykes (sites 9, 13, 14, 15).

3. Between latitude 31°50'S and 32°22'S occur gently deformed early Mesozoic volcanic breccias and flyschoid sediments that are crosscut by mafic dykes, one of which is part of this study (site 8).

Results presented in this study include paleomagne-

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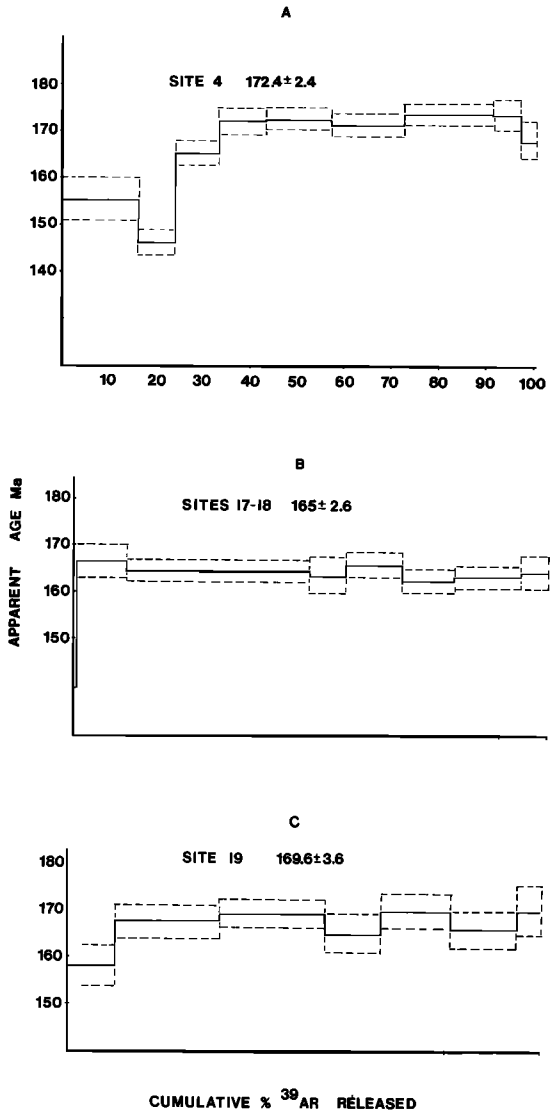


Fig. 2. $^{40}Ar-^{39}Ar$ incremental heating age spectra for hornblende from sites 4, 17-18, and 19. Age is plotted versus percentage of total ^{40}Ar released.

tism, $^{40}Ar-^{39}Ar$ and K-Ar study of the mafic dykes and fine-grained intrusion that crosscut the geologic units mentioned above, and K-Ar plus paleomagnetic data from the pluton mentioned in geologic unit 2.

Paleomagnetic Sampling

The paleomagnetic sampling sites that form the basis of this study are shown in Figure 1. Sampling includes 13 sites from mafic dykes, four sites from a quartz diorite pluton, and two sites from a fine-grained diorite intrusion. Six to 10 oriented cores were collected at each of 19 localities. Standard coring techniques were employed. Orientation of cores was done with a Brunton compass and, when conditions permitted, checked with a solar compass. No discrepancies between magnetic and solar orientations were noted. The objective of paleomagnetic sampling was to obtain specimens from a variety of datable rock units distributed over a large latitudinal range. Comparison of the paleomagnetic results from these sites with each other as well as with data from cratonal South America enables constraints to be placed on the tectonic evolution of this part of the western South American margin.

Sample Petrography

The dykes form conspicuous linear features between one and three metres wide that crosscut the older rock units with chilled margins. On the basis of their similarity in petrography and chemistry, mafic dykes 5, 6, and 7 are considered to be cogenetic. Dykes 13, 14, and 15 are correlated by the same criteria. The whole-rock chemistry data for the sampling sites of this study are presented elsewhere (J. Irwin and B. Turrin, manuscript in preparation, 1987). The K-Ar radioisotopic ages of minerals from sites 6 and 15 are considered to be representative of the three sites in each of their respective groups.

Except for sites 1 and 3, the mafic dykes involved in this study are medium- to fine-grained biotite or hornblende lamprophyres. It should be noted that lamprophyres are a distinctive type of alkalic intrusion commonly found within intracontinental rift zones such as East Africa, a topic discussed in a separate publication (J. Irwin and B. Turrin, manuscript in preparation, 1987). Alteration of igneous mineralogy and texture typically consists of chlorite in the groundmass, plus a few grains of calcite. In many of the dykes the plagioclase phenocrysts are incipiently altered to sericite plus zoisite. Site 1 consists of an ultramafic dyke initially rich in olivine and clinopyroxene, now altered to fine phyllosilicates. Site 3 is a medium-grained gabbro. Sites 10, 11, 12, and 16 are from a medium-grained, equigranular, quartz diorite pluton. Sites 17 and 18 are from a fine-grained, equigranular, diorite intrusion.

Examination of polished thin sections from sites 4, 6, 8, and 15 revealed euhedral, octahedral magnetite phenocrysts in the groundmass. Some of these phenocrysts have fine hematite rims. Within the phenocrysts of clinopyroxene and plagioclase occur elongated spindles of an opaque mineral that cannot be positively identified but which are probably magnetite.

$^{40}Ar-^{39}Ar$ Dating Results

Hornblende from sites 4, 17-18, and 19 was dated using the $^{40}Ar-^{39}Ar$ incremental heating technique. Hornblende from site 4 yields a mildly disturbed age spectrum (Figure 2) whose principal feature is a well-defined age plateau at 172.4 ± 2.4 Ma. The plateau comprises the last six increments of gas, which constitute >67% of the total ^{40}Ar released. The first three steps in this age spectrum reflect minor loss of radiogenic ^{40}Ar in the last 146 Ma, as indicated by the age of the gas released in the first few step heating increments. Hornblende from sites 17 and 18 yields an age spectrum in which all gas fractions of significant size define a plateau at 165.0 ± 2.6 Ma (Figure 2). Hornblende from site 19 yields a well defined age plateau of 169.6 ± 3.6 Ma (Figure 2). This plateau comprises six heating steps making up 90% of the total ^{40}Ar released. Very minor loss of radiogenic ^{40}Ar (<1%) may

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TABLE 1. $^{40}Ar-^{39}Ar$ Analytical Data

Step Temperature, °C	$^{40}Ar/^{39}Ar_K$	$^{37}Ar/^{39}Ar_K$	$^{36}Ar/^{39}Ar_K$	$^{39}Ar_Kr$, $\times 10^{-8}$ m^3 STP	Cumulative ^{39}Ar, %	$^{40}Ar^*/^{40}Ar$ Total	$^{40}Ar^*/^{39}Ar_K$	Apparent Age, Ma	Uncertainty, Ma
				Site 4					
800	16.05	1.50	0.311	1.01	0.163	42.7	6.850	155.4	4.7
900	8.12	3.74	0.0057	0.461	0.238	82.1	6.434	146.3	2.7
1000	8.909	13.51	0.0054	0.560	0.328	90.1	7.315	165.5	2.5
1050	8.453	12.99	0.0028	0.621	0.429	92.2	7.612	171.9	2.5
1155	8.403	14.49	0.0027	0.966	0.725	90.5	7.603	171.7	2.3
1225	8.517	15.63	0.0027	1.15	0.911	90.5	7.710	174.0	2.2
1300	9.072	17.24	0.0047	0.363	0.969	84.8	7.691	173.6	3.2
1550	10.30	17.25	0.0096	0.191	1.000	72.4	7.457	168.5	4.8
Total gas								167.1	2.2
				Site 17					
800	92.77	1.4	0.2928	0.077	0.014	6.7	6.230	140.6	8.3
1000	9.313	18.9	0.0063	0.551	0.113	80.0	7.453	166.9	3.5
1075	8.001	14.7	0.0021	2.12	0.495	92.1	7.372	165.2	2.3
1125	8.277	14.3	0.0032	0.434	0.574	88.6	7.331	164.3	3.9
1200	7.924	14.9	0.0016	0.648	0.691	93.9	7.437	166.6	2.6
1275	8.190	13.9	0.0028	0.648	0.808	89.8	7.284	163.3	2.6
1400	7.785	13.9	0.0016	0.785	0.950	93.9	7.313	163.9	2.3
1550	7.733	13.7	0.0012	0.280	1.000	95.4	7.374	165.2	3.6
Total gas								164.7	2.7
				Site 19					
800	66.86	7.4	0.1960	0.117	0.037	13.4	8.943	200.2	18.5
1000	110.89	18.5	0.0132	0.216	0.104	64.1	6.982	158.2	4.3
1075	8.818	22.7	0.0047	0.690	0.321	84.1	7.417	167.6	3.4
1125	8.558	22.2	0.0036	0.714	0.545	87.7	7.502	169.4	3.1
1175	8.868	23.8	0.0053	0.389	0.667	82.4	7.307	165.2	4.1
1250	9.218	22.7	0.0057	0.458	0.811	81.6	7.523	169.9	3.9
1350	8.796	23.3	0.0048	0.418	0.942	83.7	7.365	166.5	4.1
1550	9.381	23.3	0.0062	0.186	1.000	80.6	7.556	170.6	5.2
Total gas								168.6	3.6

be indicated by the 1000°C extraction, which yields an age of 158.2 ± 4.3 Ma, which is distinct from the plateau age, though not from all of its component steps. The $^{40}Ar-^{39}Ar$ analytical data are presented in Table 1.

The plateau ages correspond to the time at which these hornblendes cooled below their closure temperatures [*Turner,* 1968; *Dodson,* 1973, 1979]. The actual value of the closure temperature varies with geological cooling rate and the length scale of volume diffusion within individual hornblendes, but probably lies in the range 450°C-550°C [*Harrison,* 1981]. This temperature interval corresponds with the range in which the remanent magnetization of these samples was acquired: the plateau ages may be taken as close to the age of acquisition of remanent magnetization in sites 4, 17-18, and 19 (discussed below).

The close correspondence between the $^{40}Ar-^{39}Ar$ plateau ages and total gas ages (which are equivalent to conventional K-Ar ages) in the three samples analyzed with the $^{40}Ar-^{39}Ar$ technique, indicates that the conventional K-Ar ages on hornblende from the other samples in this study can with confidence be interpreted as the age of initial cooling of these igneous bodies.

K-Ar Dating Results

In order to provide information bearing on the thermal history of these samples, and as a test for the possible presence of excess argon, wherever possible, K-Ar dating was done on two separate mineral concentrations extracted from the same sample. Presented in Table 2 are descriptions of the sampling sites and discussion of the significance of the K-Ar ages. The K-Ar analytical data are presented in Table 3. Some of these samples show slight evidence of alteration, probably deuteric. It is likely that the discordance in K-Ar radioisotopic ages is related to slow uplift and cooling. Hornblende yielded the most consistent results from all of the mineral pairs.

Paleomagnetic Results

Orthogonal projections of magnetic remanence vectors [*Zijderveld,* 1967] showing response of representative specimens to thermal and alternating field (AF) demagnetization are shown in Figure 3. The behavior of these samples in thermal and AF demagnetization indicates that

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TABLE 2. K-Ar Sample Descriptions and Ages

Site	Description	Radioisotopic Age(s), Ma	Preferred Age and Discussion
1	Ultramafic dyke, olivine phenocrysts altered to phyllosilicates. Unstable during demagnetization.	None	Unknown.
2	Hornblende lamprophyre, fresh, unaltered. Stable, normal polarity RM.	Hornblende 133 Plagioclase 184	Hornblende age of 133 Ma. Plagioclase age reflects excess argon. Discordance of ages cannot be due to metamorphism:
3	Medium-grained gabbro. Unstable during demagnetization.	Hornblende 181	
4	Hornblende lamprophyre, plagioclase partially altered to zoisite + sericite, groundmass altered to chlorite + calcite. Reversed polarity site with normal overprint removed during demagnetization.	Hornblende 172	$^{39}Ar-^{40}Ar$ incremental heating data show evidence of thermal disturbance at < 146 Ma. well defined age plateau at 172 Ma. This dyke crosscuts dykes of sites 5, 6 and 7.
5,6,7	Biotite lamprophyres, plagioclase incipiently altered to sericite, biotite incipiently chloritized. Stable, normal polarity RM	Biotite 183 Plagioclase 179	Biotite age of 183 Ma. Crosscut by dyke of site 4: must be older than 172 Ma. Plagioclase and biotite K-Ar ages are concordant.
8	Biotite lamprophyre, similar to 5,6,7 above.	Biotite 144 Plagioclase 117	Biotite age of 144 Ma. Plagioclase age reflects alteration.
9	Hornblende lamprophyre, groundmass altered to chlorite + calcite. Stable, normal polarity RM.	Hornblende 151	Hornblende age of 151 Ma (no other datable minerals).
10,11,12,16	Quartz diorite pluton, mostly unaltered, localized sericite + quartz veinlets. Stable, normal polarity RM.	Hornblende 148 Biotite 154	These ages are concordant within analytical errors.
13,14,15	Hornblende lamprophyres, plagioclase locally altered to sericite. Stable, normal polarity RM.	Hornblende 134 Plagioclase 115	Hornblende age of 134 Ma. Plagioclase age reflects alteration. These dykes crosscut quartz diorite pluton above.
17,18	Fine-grained diorite intrusion, fresh. Site 17 stable, normal polarity RM. Site 18 stable, RM of both normal and reversed polarity.	Hornblende 165	Hornblende age well-defined $^{39}Ar-^{40}Ar$ incremental heating age plateau at 165 Ma.
19	Hornblende lamprophyre, fresh. Stable, reversed polarity RM.	Hornblende 170	Hornblende age well-defined, 170 Ma $^{39}Ar-^{40}Ar$ plateau.

TABLE 3. K-Ar Analytical Data

Sample and Location, South latitude, East longitude	Type	Mineral	K+ % ± Standard Deviation	Ar* mole/gram	Ar* %	Age, Ma ± 2 σ
Site 2 33°33′, 71°36′	Mafic dyke	Hb. Plag.	0.412 ± 0.01 0.122 ± 0.006	9.870×10^{-10} 4.085×10^{-11}	44 76.7	133.1 ± 7.2 183.7 ± 16
Sites 5, 6, 7 33°26′, 71°42′	Mafic dykes	Biot. Plag.	2.432 ± 0.035 1.04 ± 0.01	8.141×10^{-10} 3.319×10^{-10}	89.4 78.6	183.4 ± 6 179.4 ± 4.8
Site 8 32°18′, 71°33′	Mafic dyke	Biot. Plag.	3.416 ± 0.04 0.838 ± 0.01	8.906×10^{-10} 1.767×10^{-10}	80.3 76	144.4 ± 3.6 117.7 ± 3
Site 9 32°32′, 71°33′	Mafic dyke	Hb.	0.492 ± 0.019	1.342×10^{-10}	80.9	150.8 ± 2.2
Sites 10, 11, 12, 16 32°31′, 71°31′	Quartz diorite Pluton	Hb. Biot.	0.480 ± 0.04 6.142 ± 0.006	1.281×10^{-10} 1.709×10^{9}	73.7 87.1	148.0 ± 12 153.7 ± 0.8
Sites 13, 14, 15 33°30′, 71°33′	Mafic dykes	Hb. Plag.	0.528 ± 0.014 2.699 ± 0.035	1.276×10^{-10} 5.533×10^{-10}	65.2 93.4	134.3 ± 10 114.5 ± 4

Sites 17, 18, 19 located at 33°06′, 71°43′.
Ar* is radiogenic argon of mass 40.

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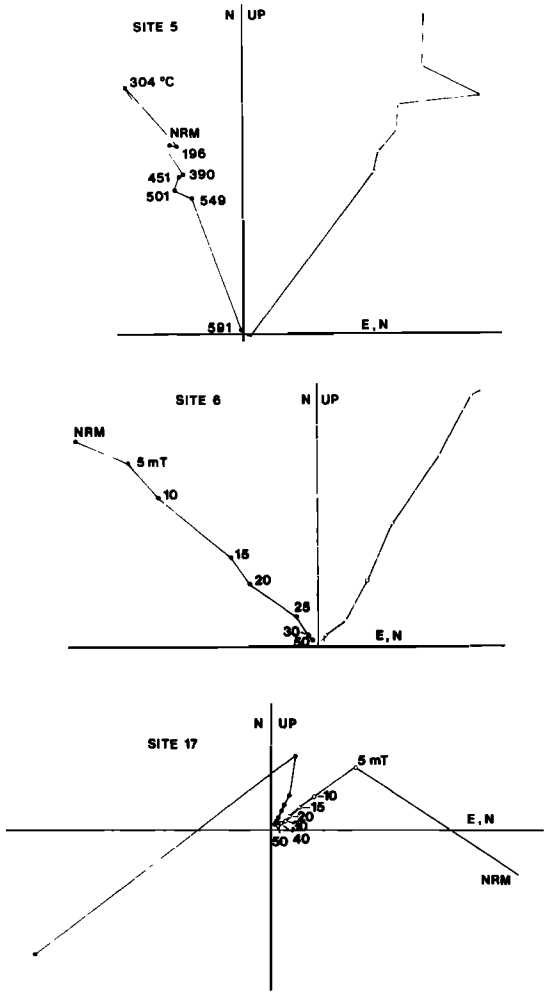


Fig. 3. Orthogonal projection of remanence vector during AF and thermal demagnetization of representative specimens. Closed (open) circles projected onto horizontal (N-S vertical) plane. Tick marks on axes are amperes/meter.

magnetite is the principal carrier of remanent magnetization (RM). Figure 4 shows typical normalized remanent magnetization intensity behavior during thermal demagnetization. The acquisition of isothermal remanent magnetization behavior of some of these specimens is shown in Figure 5. The saturation level of less than 0.1 Tesla (T) observed in these specimens is typical of RM carried by magnetite. Undemagnetized natural remanent magnetization (NRM) directions are well grouped in seven of 16 sites with northward declinations and upward inclinations. Figure 6 illustrates the distribution of magnetically cleaned mean RM directions from all sites. As illustrated by orthogonal projections of AF and thermal demagnetization of representative specimens (Figure 3), most of the specimens in this study carried single-component RM. A soft, secondary component of RM, readily eliminated by AF and thermal cleaning, was detected during demagnetization of a few specimens from site 4. Site 17 in Figure 3 shows what appears to be removal of a soft viscous component. Two sites (1 and 3) did not respond to either thermal or AF demagnetization and were abandoned. The plutonic rocks (sites 10-12, 16-18) have low values of median destructive fields, behavior typical of plutons [*Beck*, 1980].

Figure 7 shows paleomagnetic poles and 95% confidence intervals [*Fisher*, 1953] calculated by averaging the VGP's from sites of similar age (listed in Table 4). The Jurassic paleomagnetic pole for cratonic South America is not distinguishable from the present rotation axis, and as a result, NRM directional data alone cannot distinguish a primary Jurassic direction from a Quaternary secondary magnetization. Characteristics of these samples that are consistent with a primary origin of their characteristic RM include the following: (1) These specimens yielded a tightly clustered component of RM of both normal and reversed polarity. (2) No evidence of postemplacement reheating or chemical alteration was detected in either petrographic examination or in the $^{40}Ar-^{39}Ar$ incremental heating experiments. (3) Most of the mafic dykes in this study have similar W-NW trends and near-vertical dips (Figure 8). The two mafic dyke sites of reversed polarity (which contain RM antipodal to the other sites in the same age grouping) come from dykes that strike N-S and dip steeply west. Any major structural correction to dyke orientations will result in a decrease in the clustering of the paleomagnetic data. It can be inferred that the dykes have not been folded in the time period after the RM was acquired. (4) Samples were collected from igneous bodies that vary widely in form and orientation, that crosscut a variety of rock units and from sites distributed over a large latitudinal range. It is unlikely that tilting could have affected systematically such a large area and variety of rock units. (5) One of the sampling sites (site 8) is a steeply dipping mafic dyke that crosscuts flat lying Mesozoic sediments. The mean direction from this site is consistent with the other coeval site mean directions. None of the observations outlined above is in itself sufficient to establish the primary nature of the characteristic NRM. However, taken together, these observations provide strong evidence that the characteristic NRM is primary.

No detailed magnetostratigraphy has yet been published for the timespan of this study (183-134 Ma). The best paleomagnetic data indicate a predominance of nor-

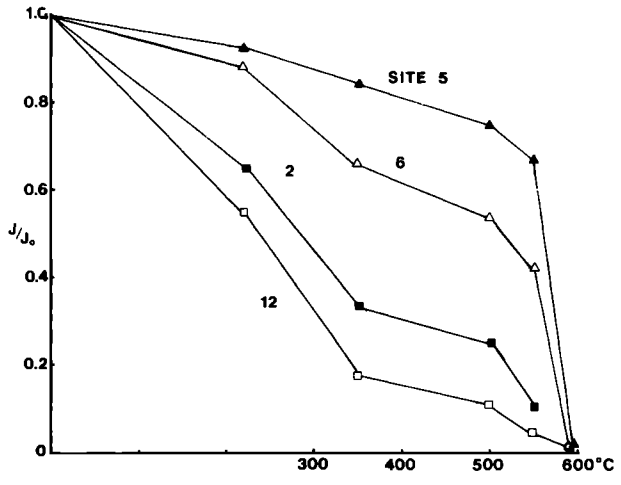


Fig. 4. Normalized RM intensity of representative specimens in thermal demagnetization.

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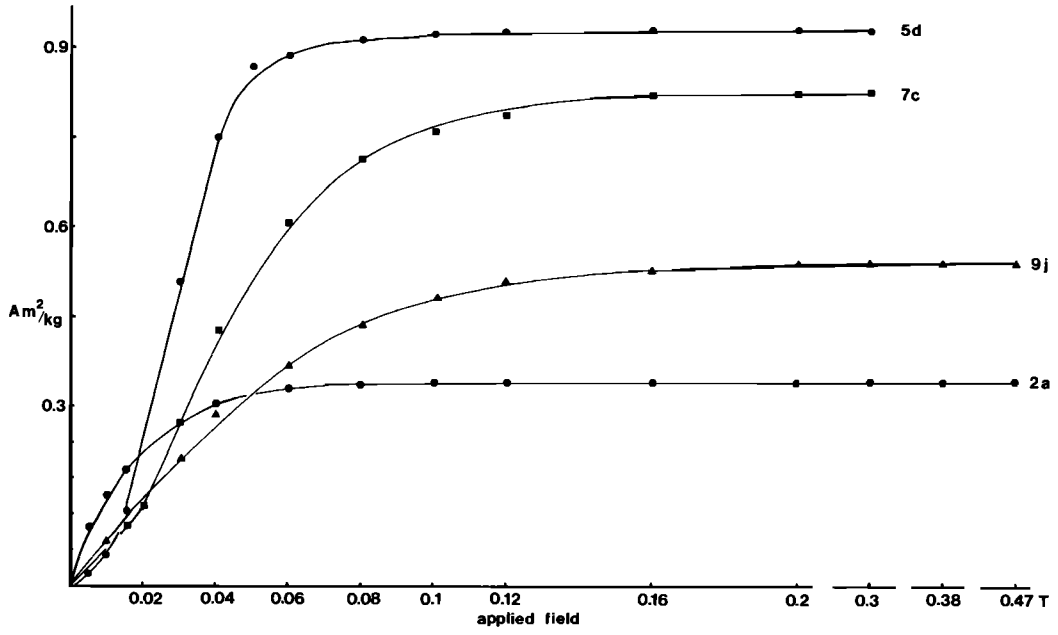


Fig. 5. Acquisition of isothermal remanent magnetization in representative specimens.

mal polarity during the Jurassic. In the very large data set from the Newark trend intrusions compiled by *Smith and Noltimier* [1979], not a single reversed polarity site is reported from a total of 156 sites dated at 179 Ma. Similarly, *Dalrymple et al.* [1975] encountered only six reversed sites out of 25 sites dated at 180 ± 10 Ma, and *Kluth et al.* [1982] report five reversed sites out of a total of 15 dated by K-Ar at close to 150 Ma. Two sites in this study (4 and 19) have reversed RM directions antipolar to the mean of the other sites in the same age group. Mixed polarity occurs at site 18: four cores of normal polarity and two of reversed. *Harland et al.* [1982] and *Kent and Gradstein* [1985] show both normal and reversed polarity epochs, with a predominance of normal polarity, for the time period of 183-134 Ma. Field reversals have been reported in the Kimmeridgian to Tithonian stages of the Jurassic, which possibly correspond with the age of some of the samples of this study [*Ogg et al.*, 1984]. Unfortunately, the radioisotopic ages of the geomagnetic reversals in the Jurassic are very poorly constrained—the errors on the absolute ages of the biostratigraphic stages of the Jurassic are typically 15-20 Ma [*Geological Society of America*, 1983]. It is not possible to correlate the radioisotopic ages presented here with biostratigraphically dated field polarities.

Relationship Between Radioisotopic and Paleomagnetic Data

Although petrographic and radioisotopic data presented here indicate that the thermal history of a few of these samples is not as simple as (for example) that of recent volcanics, it should be emphasized that most of these rocks do not possess significant secondary components of RM and that the $^{40}Ar-^{39}Ar$ incremental heating age spectra are typical of rocks that have relatively simple thermal histories. The relationship between the Ar retention behavior of the K-Ar and $^{40}Ar-^{39}Ar$ radioistopic systems and the blocking temperature of remanent magnetization is discussed by *Berger and York* [1979] and *Buchan et al.* [1977] and theoretically evaluated by *York* [1978]. The retention of argon in minerals and the acquisition of TRM in magnetite can be expressed by equations that have similar form [*York*, 1978]. The temperature ranges over which hornblende becomes closed to argon diffusion (550°-450° C) and fine-grained magnetite acquires TRM are similar. This means that the component of RM carried by fine-grained magnetite in igneous rocks can be

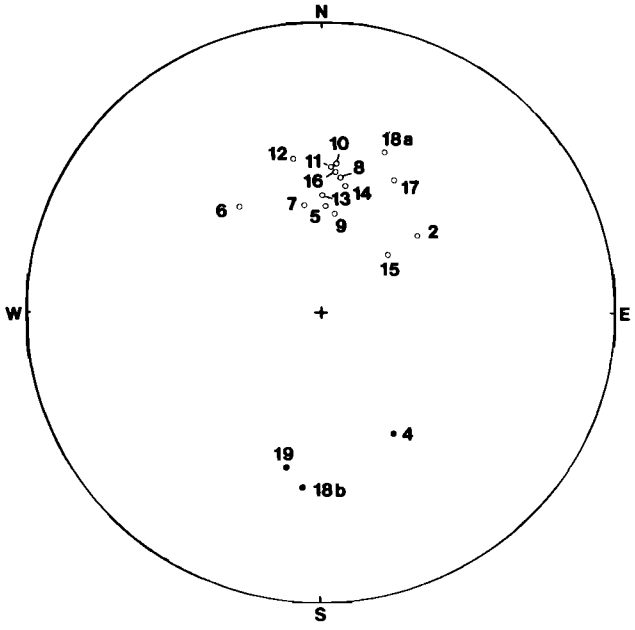


Fig. 6. Demagnetized site mean directions listed in Table 3. Open (closed) circles are upper (lower) hemisphere.

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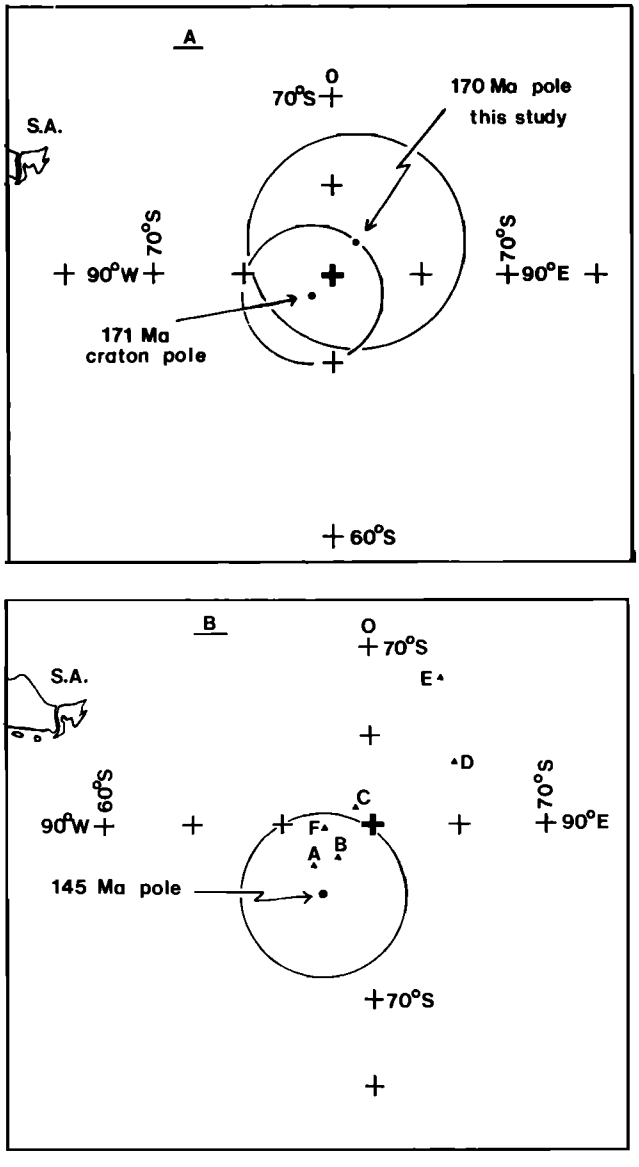


Fig. 7. Paleomagnetic poles and 95% confidence intervals [*Fisher*, 1953] calculated from VGP's. (a) Pole from 170 Ma mean (sites 4, 5, 6, 7, 17, 18a, 18b, 19) and circle of 95% confidence, triangle is average Middle Jurassic cratonal pole and 95% confidence interval [*Fisher*, 1953] calculated from references cited in text. (b) Pole from 145 Ma mean (sites 2, 8, 9, 10, 11, 12, 13, 14, 15, 16) and circle of 95% confidence. Also shown are some Cretaceous cratonal poles: A, *Opdyke and Macdonald* [1973]; B, *Valencio et al.* [1977]; C, *Schult and Guerreiro* [1980]; D, *Creer* [1962]; and E, *Mendia* [1978].

radioisotopically dated by application of the $^{40}Ar-^{39}Ar$ technique to hornblende. The older radioisotopic age of each discordant mineral pair, and the broad plateau in the $^{40}Ar-^{39}Ar$ age spectra are close to the age of acquisition of the characteristic RM.

The NRM of the samples of this study is carried mostly by magnetite. Octahedral magnetite phenocrysts occur within the groundmass in some of these samples and as inclusions in clinopyroxene and plagioclase phenocrysts. Magnetite is not stable in the range of conditions of oxygen fugacity typical of low-grade metamorphism or near-surficial alteration [cf. *Carmichael et al.*, 1974, p. 331, *Hyndman*, 1985, p. 496], and except under conditions of unusually low oxygen fugacity usually manifested by the presence of reduced sulphide or carbon compounds—none of which are present in the samples of this study—magnetite is not produced in the weathering environment [e.g., *Palache et al.*, 1944]. In the absence of any evidence that magnetite was produced by postintrusion alteration and having observed igneous phenocrysts of magnetite in polished thin section, it is clear that the magnetite is a primary igneous mineral, the age of cooling of which is measured by the radioisotopic ages. The fine hematite rims present on some of the euhedral magnetite phenocrysts in the groundmass of these samples were most likely produced deuterically during postintrusion cooling or perhaps by later weathering. The contribution of hematite to the measured RM is negligible.

In summary, although no specific field tests constrain the timing of acquisition of RM, the petrographic and isotopic characteristics of these samples indicate a simple enough thermal history that it is unlikely that the RM carried by igneous magnetite phenocrysts has been thermally reset. The relatively straightforward behavior of most of these specimens during demagnetization is consistent with a primary origin of their RM. The time of acquisition of this RM is probably close to the radioisotopically determined ages—particularly hornblende.

TECTONIC INTERPRETATION

Virtual geomagnetic poles from each site are listed in Table 4. The 17 stable sites were divided into two groups at 170 and 145 Ma on the basis of the radioisotopic data. Paleomagnetic pole directions (illustrated in Figure 7 and listed in Table 5) were recalculated by averaging the VGP's from each of the two age groups. Paleomagnetic poles from the Chon Aike Formation in southern Argentina [*Valencio and Vilas*, 1970; *Creer et al.*, 1972; *Vilas*, 1974] and from the Porto Franco volcanics [*Schult and Guerreiro*, 1979] were averaged to determine the Middle Jurassic South American cratonal paleopole for comparison with the 170 Ma pole from this study. The ages of Chon Aike Formation lavas were determined by conventional K-Ar age dating of phyllosilicate mineral separates [*Creer et al.*, 1972]. When recalculated using the revised constants of Steiger and Jaeger [1977], the average age of the lavas sampled in the Chon Aike Formation (listed by *Creer et al.* [1972]) is 171 Ma. K-Ar ages of the Porto Franco volcanics average betwen 158 Ma and 160 Ma [*Schult and Guerreiro*, 1979]. A mean cratonal South American pole calculated by averaging the data from the Chon Aike Formation and the Porto Franco volcanics coincides with the 170 Ma pole of this study (Figure 7*a* and Table 5). A considerably greater degree of scatter—the source of which is unclear, is evident in the Jurassic to Cretaceous paleopole directions from South America. *Valencio et al.* [1983] propose that several sharp bends occur in the South American apparent polar wander path during the Jurassic and Cretaceous. Unfortunately, there are no well-dated 155—140 Ma cratonal poles to be compared with the 145 Ma pole of this study. Having established that the pole from older rocks (170 Ma) is consistent with coeval poles from the craton, however, it is

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TABLE 4. Paleomagnetic Site Direction Data

Site	Lithology†	Demag-netization	N	R	k	a 95	Decli-nation	Incli-nation	Age, Ma	VGP	
										South Latitude	East Longitude
1	md	unstable during demagnetization									
2	md	35 mT	6	5.86	36	11.4	51.8	−55.1	133	47.8	216.6
3	md	unstable during demagnetization									
4	md	35 mT	6	4.88	42	10.5	150.2	49.6	172	*64.5	16.8*
5*a*	md	549°C	6	5.84	30.9	12.1	345.4	−55.7	183	*77.4	1.3*
5*b*	md	25 mT	7	6.96	138	4.5	344.7	−57.8	183	*76.8	351.6*
6	md	30 mT	10	9.9	92	5.1	330.8	−52.7	183	65.9	10.6
7	md	40 mT	9	8.91	91	5.4	350.7	−59.3	183	80.2	333
8	md	45 mT	8	6.97	198	4.3	7.8	−50.1	144	83.2	188.4
9	md	45 mT	5	8.96	191	3.7	8.1	−61.4	151	78.1	258.2
10	pl	25 mT	7	7.79	34	9.6	5.7	−46.8	154	83.3	157.4
11	pl	25 mT	6	4.75	16	19.8	4	−47	154	84.5	148.2
12	md	25 mT	6	6.58	14	16.5	350.2	−45.7	154	79.8	47.6
13	md	50 mT	6	5.99	674	2.6	0.8	−58.2	134	83.6	282.9
14	md	40 mT	4	5.94	82	7.4	10.7	−53.2	134	81	209.3
15	md	30 mT	6	5.88	40	10.7	49.8	−64.9	134	49.8	234.4
16	pl	20 mT	6	3.93	45	13.9	5.8	−48.2	154	84	166.6
17	fgi	30 mT	6	5.79	23.8	11.7	28	−47.1	165	65.5	194.5
18*a*	fgi	25 mT	4	3.92	37.5	9.3	21.2	−41.4	165	69.2	177.2
18*b*	fgi	30 mT	2	1.99	100	9.9	186	40	165	78.4	136.9
19	md	15 mT	6	5.94	83.3	6.3	192.4	44.4	170	78.8	54.6

† md_r, mafic dyke; pl, plutonic; fgi, fine-grained intrusion.
* Site 5 average VGP = 77.2, 35.

likely that the 145 Ma pole should be coincident as well. In Figure 7*b* is shown the 145 Ma pole of this study and examples of Cretaceous poles from the South American craton. The scatter in the cratonal data is presumably due to the variation in their age. Most of these paleomagnetic studies are dated by the K-Ar whole-rock method. Samples that have been subjected to any postcooling hydrothermal alteration or weathering are subject to argon loss; therefore whole-rock K-Ar ages must be considered as minimum (and probably unreliable) ages for cooling of Mesozoic igneous rocks. The Jurassic and Cretaceous South American cratonal poles are not yet sufficiently constrained in age to permit further editing of this data set.

SUMMARY

Coeval sites distributed over a large area of the coastal cordillera of Chile possess a stable component of RM isolated during AF and thermal demagnetization that is consistent between the sites involved in this study and consistent with published poles for cratonal South America.

CONCLUSIONS AND DISCUSSION

Paleomagnetic data from the coastal cordillera of Chile (latitude 32°25'-33°35'S) indicate that no major translation or large rotations of crustal blocks have affected this part of the continental crust since Middle Jurassic time.

Some of the geological characteristics of central Chile are consistent with an autochthonous late Mesozoic and Cenozoic evolution of this region [*Dalziel and Forsythe,* 1985]. All of the tectonostratigraphic terranes in western North America that are demonstrably allochthonous occur "outboard" (toward the paleo-Pacific ocean) of former plate margins [e.g., *Monger et al.,* 1972]. No rock assemblages characteristic of a former plate boundary have been reported in the Mesozoic and Cenozoic of central western South America. If allochthonous terranes

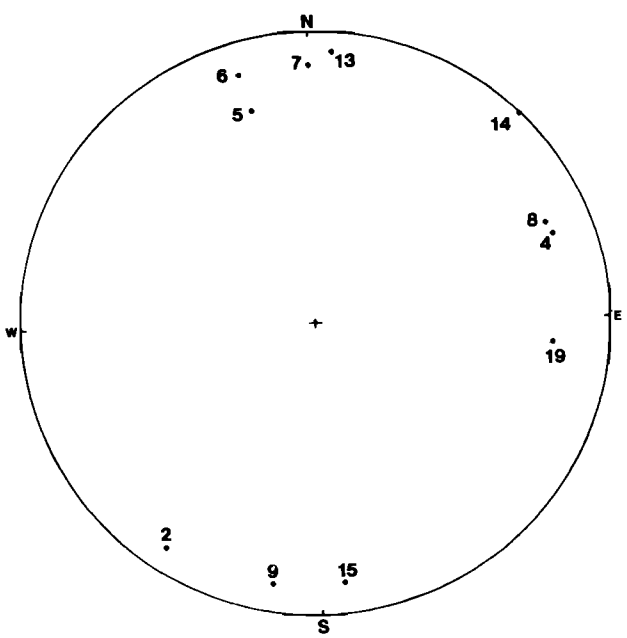


Fig. 8. Mafic dyke orientations (poles to planes shown), numbered according to paleomagnetic site number.

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TABLE 5. Paleomagnetic and South American Cratonal Poles

Age Group	Paleomagnetic Poles						
	Mean Declination	Mean Inclination	N	Average VGP Latitude	Average VGP Longitude	a 95	k
170 Ma	2	-49.2	8	85.9°S	36.7°E	11.9	17.4
145 Ma	11.4	-54.3	10	80.3°S	207.8°E	9.5	21.58

K-Ar Age	South American Cratonal Poles: Pole Latitude, Longitude	Reference
75 Ma*	82°S, 127°W	*Opdyke and Mcdonald* [1973]
85-99 Ma*	87.6°N, 135°E	*Schult and Guerreiro* [1980]
114 Ma*	85°S, 222°E	*Valencio et al.* [1977]
123 Ma*	72°S, 25°E	*Mendia* [1978]
Lower Kt	78.2°, 125.9°W	*Creer* [1962]
160 Ma	85.3°N, 82.5°E	*Schult and Guerreiro* [1979]
171 Ma	85°S, 163°W	*Vilas* [1974]
171 Ma	82°S, 129°W	*Creer et al.* [1972]
Mid Ju	84°S, 56°E	*Valencio and Vilas* [1970]

†Mean Middle Jurassic pole from these studies is 86.7°S, 228.9°E.

*K-Ar whole-rock ages and probably minimum estimates of the age of cooling.

have been accreted to the central western South American margin during the late Mesozoic or Cenozoic, then the geologic evidence of such, in the form of accretionary subduction complexes, has not been recognized. Transform faults such as the San Andreas and the Alpine fault of New Zealand have characteristic geologic signatures. No examples of these features have been documented in central western South America.

Geologists working in western South America have suggested that extensional tectonics dominate the late Mesozoic and Cenozoic evolution of the Andean cordillera in central Chile [*Katz,* 1971; *Drake,* 1982; *Aberg et al.,* 1984]. One would not expect extension to be the dominant structural feature if central Chile had been a zone of collision between buoyant continental or "microcontinental" plates brought together by closure of an ocean basin.

The origin of the mid-Paleozoic high-grade metamorphic complex described previously remains unknown; it may have been added to western South America during an earlier stage in the evolution of this plate margin. Blueschist facies metamorphic rocks dated at 329 Ma occur within melange at Pichilemu, 100 km to the south of this study area on the coast [*Hervé et al.,* 1974]. The character and evolution of the Paleozoic metamorphic series of coastal Chile are little studied and may hold critical elements of the history of this long-lived zone of dynamic plate interaction.

The paleomagnetic results presented here are consistent, or at least easily reconciled, with other published paleomagnetic studies of central Chile [*Beck et al.,* 1986]. In their study of Cretaceous ash flows that occur between the coastline and the Andean cordillera of central Chile, *Beck et al.* [1986] present paleomagnetic and radioisotopic data (albeit of a preliminary nature) that indicate no rotation or translation for several Cretaceous sampling localities 100 km inland from the coast at the same latitude as the area of the Jurassic and Cretaceous sampling sites of this study (Chacabuco tuffs of *Beck et al.* [1986]). Study of ash flows from a locality 100 km to the south of the southernmost sampling site of this study (and which are 200 km south of the Chacabuco tuffs) appear to yield modest clockwise rotations (14°) in paleomagnetic declination, relative to an imprecisely dated Cretaceous cratonal reference pole. As shown on the Mapa Geologica de Chile [*Instituto de Investigaciones Geológicas,* 1980], the regional strike of lithologic and structural trends in central Chile changes in the area immediately to the south of the paleomagnetic sampling sites of this study. The sense and magnitude of deflection in the strike of these regional lithologic trends and the change in the strike of the coastline corresponds exactly with the modest clockwise rotation of paleomagnetic declinations observed at the San Fernando site of *Beck et al.* [1986].

A possible mechanism for the observed change in regional lithologic trends and paleomagnetic declinations is readily apparent from examination of the Mapa Geologica de Chile [*Instituto de Investigaciones Geológicas,* 1980] and the distribution of radioisotopically dated igneous rocks in central Chile shown in *Drake et al.* [1982]. Large volumes of Neogene and Quaternary igneous rocks have been intruded into the crust of central Chile only in the area south of latitude 33°20'S, which corresponds with the area of the observed rotations. Igneous rocks of this age are sparse in the unrotated area to the north. Crustal extension related to magmatism is a likely cause of the rotated paleomagnetic declinations reported at the San Fernando site of *Beck et al.* [1986]. Modest rotations of paleomagnetic declinations, related to crustal extension, are also observed in the Afar triangle of eastern Africa [*Courtillot et al.,* 1984].

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APPENDIX: ANALYTICAL METHODS

K-Ar Dating

All sample preparation, potassium analyses, argon extraction, and measurement were done in the Department of Geology and Geophysics at the University of California at Berkeley. Potassium analyses were made using a phase modulation (PM) 5 flame photometer on splits from samples fused for argon extraction. Argon analyses were done on a Reynolds type mass spectrometer using the isotope dilution technique described by *Dalrymple and Lanphere* [1969].

$^{40}Ar/^{39}Ar$ *Incremental Heating Analyses*

Hornblende separates were prepared using conventional techniques followed by hand picking under a binocular microscope. Grain sizes ranged from 100 to 200 microns. Irradiation was carried out in the High Flux Beam Reactor at Brookhaven National Laboratory. Production of neutron-induced isotopes was monitored using MMhb-1 [*Alexander et al.,* 1978] and pure K and Ca salts. Factors used to correct the effects of interfering neutron reactions were $(^{40}Ar/^{39}Ar)K = 1.25 \times 10^{-2}$, $(^{36}Ar/^{37}Ar)Ca = 2.24 \times 10^{-4}$, $(^{39}Ar/^{37}Ar)Ca = 6.64 \times 10^{-4}$. Argon was extracted using an RF induction furnace [*Schaeffer and Schaeffer,* 1977] and purified using Ti and Zr-Al getters. Procedural blanks ranged from 1 to 5 $\times 10^{-9}$ cm^3 ^{40}Ar in the course of the study. Argon analyses were performed using a 20 cm radius 60° sector glass mass spectrometer operated in the static mode at the State University of New York, Stony Brook. Mass discrimination was monitored before and after each sample using an air pipette and was 0.80 % /amu.

As is typical of Mesozoic or younger hornblende samples with low K+ contents (0.17-0.32), uncertainties in apparent ages result largely from the correction of the ^{40}Ar signal for ^{40}Ar-air based on ^{36}Ar-air. Apparent ages were calculated using the constants of *Steiger and Jaeger* [1977]. Uncertainties quoted for apparent ages are 1 σ and include contributions from blank corrections, monitor parameters, neutron-produced interferences and signal measurements.

Paleomagnetism

Paleomagnetic data were collected on both spinner and cryogenic magnetometers at the laboratories listed in the acknowledgments. At least two specimens and, in most cases, all of the specimens from each site were progressively AF demagnetized in 5-mT intervals in peak fields up to 80 mT. The remaining specimens from each site were then stepwise cleaned in optimum fields determined from the demagnetization plots. All specimens were demagnetized in four or more increments. At least one specimen from each site was incrementally thermally demagnetized up to 550°C or 591°C, depending on the behavior of the sample. In order to compare the behavior of samples in thermal and AF demagnetization, six specimens from site 5 were progressively thermally demagnetized up to 591°C. The site mean calculated from thermal demagnetization (at 549°C) of these specimens overlaps with the site mean calculated from AF demagnetization (at 25 mT) of the entire site (Table 5). The characteristic site mean demagnetization level (listed in Table 5) was selected on the basis of inspection of orthogonal projections of the RM vector during stepwise demagnetization and by determination of the demagnetization level at which the RM vector is statistically most tightly clustered.

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